

HW 6

4.2

The probability of getting a single pair in a poker hand of 5 cards is approximately 0.42. Find the approximate probability that out of 1000 poker hands there will be at least 450 with a single pair.

✓ Answer ✓

We may approximate this as normal with the parameters of $N(np, \sqrt{np(1-p)})$.

```
use stats::{distribution::{Binomial, ContinuousCDF, DiscreteCDF,
Normal}}, statistics::Distribution};

fn q402() {
    let p = 0.42;
    let n = 1000;
    let dist = Binomial::new(p, n).unwrap();
    let approx = Normal::new(dist.mean().unwrap(),
dist.std_dev().unwrap()).unwrap();
    println!("Actual Binomial: {}", 1f64 - dist.cdf(450-1));
    println!("Normal Approximation: {}", 1f64 - approx.cdf(449.5));
}
```

By utilizing this code we achieve an estimate for this answer:

Actual Binomial: 2.9638126591587266%

Normal Approximation: 2.9372703289962887%

4.6

A pollster would like to estimate the fraction p of people in a population who intend to vote for a particular candidate. How large must a random sample be in order to be at least 95% certain that the fraction $\hat{p}$ of positive answers in the sample is within 0.02 of the true $\hat{p}$?

✓ Answer

Assuming the distribution is roughly normal,

$$ME = z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

$$n = \left(\frac{z_{\alpha/2}}{ME}\right)^2 p(1-p)$$

We can maximise p by setting it to 0.5 to provide the most conservative estimate for n .

By rounding up, this provides us with a conservative estimate of 2401.

4.10

A hockey player scores at least one goal in roughly half of his games. How would you estimate the percentage of games where he scores a hat-trick (three goals)?

✓ Answer

If this player scores at least one goal for half of his games, then approximating with Poisson, our λ should be $\ln 2$.

This means that the 3 or more goals would be represented by $P(\text{Poisson}(\ln 2) \geq 3)$

Which is approximately equal to 3.331315624047693% of games.

4.16

We choose 500 numbers uniformly at random from the interval $[1.5, 4.8]$.

a

Approximate the probability of the event that less than 65 of the numbers start with the digit 1.

✓ Answer

$X \sim \text{Uniform}(1.5, 4.8)$

$p = P(X < 2)$

$N \sim P(\text{Binomial}(p, 500))$

$P(N < 65)$

$\approx 7.784645720062316\%$

```
fn q416a() {
    let p = 0.5 / (4.8-1.5);
    let trials = 500;
    let n = Binomial::new(p, trials).unwrap();
    println!("{}", n.cdf(64));
}
```



b

Approximate the probability of the event that more than 160 of the numbers start with the digit 3.

✓ **Answer**

$$p = P(X \in [3, 4))$$

$$N \sim P(\text{Binomial}(p, 500))$$

$$P(N > 160)$$

$$\approx 19.05590222403083\%$$

```
fn q416b() {  
    let p = 1.0 / (4.8-1.5);  
    let trials = 500;  
    let n = Binomial::new(p, trials).unwrap();  
    println!("{}", 1.0 - n.cdf(160));  
}
```



4.24

We roll a fair die repeatedly and keep track of the observed frequencies of the outcomes $1, 2, \dots, 6$.

a

Show that the probability of seeing at least 17% fours converges to zero as the number of rolls tends to infinity.

✓ **Answer**

$$p = \frac{1}{6}$$

$$\hat{p}_n = \frac{\text{Binomial}(p, n)}{n}$$

$$P_n = P(\hat{p}_n \geq 0.17)$$

We can approximate p_n with the normal distribution

$$\begin{aligned} p'_n &\sim \frac{N(np, \sqrt{np(1-p)})}{n} \\ &= N\left(p, \sqrt{\frac{p(1-p)}{n}}\right) \end{aligned}$$

Therefore $P(p'_n > 0.17) \rightarrow 0$ as well

b

Let A_n be the event that after n rolls, the frequencies of all six outcomes are between 16% and 17%. Show that for a large enough number of rolls, the probability of A_n is at least 0.999.

✓ **Answer**

$$A_n = P(0.16 < p'_n < 0.17)$$

Since as $n \rightarrow \infty$, $p'_n \rightarrow N(p, 0)$

$$P(0.16 < p'_{n \rightarrow \infty} < 0.17) \rightarrow 1$$

For $n = 10^{500}$

$$P(0.16 < p_n < 0.17) \approx 1$$

On a 64-bit floating point computer, our STD is 0 and the computer incorrectly states that our probability *is* 1.

STD:

[illegible]
$$P(0.16 < p_n < 0.17): 1$$

```
fn q424b() {  
    let p = 1.0 / 6.0;  
    let std = (p*(1.0-p)/10.0_f64.powf(100.0)).powf(0.5);  
    println!("STD: {}", std);  
    let pn = Normal::new(p, std).unwrap();  
    println!("P(0.16 < pn < 0.17): {}", pn.cdf(0.17) - pn.cdf(0.16))  
}
```

4.26

100 randomly chosen individuals were interviewed to estimate the unknown fraction $p \in (0, 1)$ of the population that prefers whole milk to skim milk. The resulting estimate is $\hat{p}$. With what level of confidence can we state that the true p lies in the interval $(\hat{p} - 0.1, \hat{p} + 0.1)$?

✓ **Answer**

Sampling from a population with p who prefer whole milk to skim milk with a size of 100 gives us the following distribution:

$$S \sim \text{Binomial}(p, 100)$$

$$\approx N\left(p, \sqrt{\frac{p(1-p)}{100}}\right)$$

The most conservative estimate for our STD. would be at $p = 0.5$:

$$\sigma = \frac{1}{20}$$

$$2P\left(Z < \frac{20}{10}\right) \approx 95.4\%$$

So we are 95% confident.

4.35

Every morning Jack flips a fair coin ten times. He does this for an entire year. Let X denote the number of days when all the flips come out the same way (all heads or all tails).

a

Give the precise expression for the probability $P(X > 1)$.

✓ **Answer**

Probability of all same is $\frac{1}{2^9}$

Distribution of days in a year would be $X \sim \text{Binomial}(2^{-9}, 365)$.

$$P(X > 1) = 16.01997510861235\%$$

(Code provided at the next question)

b

Apply either the normal or the Poisson approximation to give a simple estimate for $P(X > 1)$. Explain your choice of approximation.

✓ **Answer**

Since p is small, we should use a Poisson approximation for X with $\lambda = np$.

$$X' \sim \text{Poisson}(365(2^{-9}))$$

$$\begin{aligned} P(X' > 1) \\ = 16.02980383532987\% \end{aligned}$$

Which is fairly close to our actual answer

```
fn q435() {  
    let p = 2_f64.powf(-9.0);  
    let b = Binomial::new(p, 365).unwrap();  
  
    println!("More than 1 days: {}", 1.0 - b.cdf(1));  
  
    let n = Poisson::new(b.mean().unwrap()).unwrap();  
    println!("Poisson Approx: {}", 1.0 - n.cdf(1));  
}
```

